

EXPERIMENTAL INVESTIGATION AND TAGUCHI-BASED OPTIMIZATION OF CHATTER CHARACTERISTICS AND MATERIAL REMOVAL RATE IN THIN- WALLED MILLING USING TIME- FREQUENCY ANALYSIS

Article history

Received

X Month 202Y

Received in revised form

XA Month 202Y

Accepted

XB Month 202Y

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Graphical abstract



Abstract

High strength aluminium alloys are used in thin-walled applications in the aerospace, automotive and lightweight structure manufacturing industries because of their important strength to weight ratio. Lower structural rigidity of thin wall part produces chatter vibration during thin wall milling, which leads to poor surface integrity, dimensional error, and endangering milling process. Proposed research aims to perform an experimental analysis and understand the chatter nature and material removal characteristics in thin-walled milling of AA7075-T6 aluminium alloy based on Taguchi L27 orthogonal array design. Material removal rate (MRR), kurtosis index and chatter frequency were tested with spindle speed, feed rate and radial depth of cut on the systematic effects and were analysed from time domain and frequency domain. Vibration signals obtained in manufacturing operations were processed to grasp the value of kurtosis as an indicator of higher order of statistical chatter, and the chatter frequency response was processed to find the dynamic instability trend with different machining conditions. Optimization of MRR subject to machining parameters and their significance was done by utilising Taguchi S/N ratio, regression modelling and ANOVA. Experimental results showed that the radial depth of cut had the most significant impact on MRR followed by feed rate and spindle speed. The optimum conditions, were employed during the verification testing, yielding a maximum MRR with a deviation of 0.98% between the experimental and prediction results. According to the kurtosis analysis, the vibration fluctuation of the Z direction is relatively large, indicating that its chatter is serious. Frequency domain analysis also revealed the chatter frequency became higher with rising of spindle speed, which meant that the dynamic excitement was raised. At unstable cutting conditions, FESEM of the machined surfaces has revealed the presence of chatter-induced defects like irregular feed marks, adhered debris, plastic deformation, and smearing on the machined surface.

Keywords: Thin-walled milling, Al7075-T6, Chatter vibration, Kurtosis index, Taguchi optimization, Material removal rate (MRR).

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1.0 INTRODUCTION

One of the most widely used high-strength alloys in aerospace, automotive, and defence industries due to its high-strength-to-weight ratio, good fatigue behaviour and

very high machinability is aluminium alloy AA7075-T6. There has been a massive transition towards lightweight materials in the global manufacturing environment due to the need to conserve energy, reduce emissions, and sustainability requirements, particularly in aerospace and electric vehicle applications. Therefore, AA7075-T6 remains a subject of much research, especially in advanced machining and hybrid processing, as well as optimization of surface integrity. Recent advancements have been made in the areas of better machinability, reduction in chatter and improvement in the quality of surfaces through optimization of process parameters and use of real-time monitoring.

An example is the use of high-speed machining and tool geometry to enhance cutting stability and reduce defects caused by vibration in milling processes [1], whereas others have studied how signal processing techniques (kurtosis and spectral analysis) can be used to detect chatter in milling processes [2]. Also, studies have noted the need to have multi-objective optimization strategies, such as Taguchi and hybrid, to strike a balance between productivity and surface integrity in machining AA7075 components [3]. Emerging technologies of sustainable and smart manufacturing have only intensified the focus on sensor-based monitoring and data-driven models to predict machining performance in different cutting conditions [4]. In addition, the effect of process dynamics and material removal mechanisms on the stability of thin-walled structures, which are naturally prone to chatter because of reduced stiffness, has also recently been studied [5].

Despite these developments, there still exists difficulties in obtaining a strong correlation between machining parameters, dynamic response, and performance measures like material removal rate and vibration characteristics, hence necessitating further experimental studies. Thin-walled milling is a critical manufacturing process that has been widely used in the aerospace, automotive and precision engineering industries where structural components are designed with minimal weight and high stiffness and strength. Thin-walled structures are however highly prone to regenerative chatter, forced vibrations and dynamic instability during machining and this significantly impairs the surface integrity and dimensional accuracy. This further complicated dynamic behaviour of thin-walled components because, in addition to decreasing stiffness as material is removed, the system dynamics varies over time, challenging traditional stability prediction models.

Recent studies have been aimed at comprehending and alleviating these effects based on the advanced modelling and experimental research works. To illustrate this, the effects of tool workpiece interaction and process damping on the stability of chatter has been studied extensively in order to enhance cutting performance [6]. The importance of the tool path strategies and adaptive machining techniques in minimizing deformation and vibration has been also shown [7]. Moreover, research has also pointed to the fact that spindle speed, feed rate, and depth of cut have a substantial effect on the amplitude of vibration and the onset of chatter and requires systematic optimization strategies to achieve stable machining [8]. It has become apparent that combining stability lobe diagrams with the real-time monitoring is an effective way to choose the chatter-free cutting parameters [9]. Moreover, signal processing methods (wavelet transforms and higher-order statistics) have become widely used in the detection and classification of chatter accurately [10].

Gurdal et al. have stated that dynamic modelling and stability prediction are critical in milling of flexible structures, with adequate identification of the system being the key to chatter suppression [11]. Budak et al. have shown that process damping and tool geometry can have a significant bearing on stability limits in thin-wall machining, especially at low cutting speeds [12]. According to Tamas et al., the stability lobe diagrams that are coupled with experimental validation are good guidelines that can be followed to avoid chatter in milling processes [13]. Croppi et al. studied the impact of workpiece flexibility and fixturing conditions and found that structural stiffness is the dominant factor in vibration suppression [14]. Stavropoulos et al. studied data-driven and sensor-based techniques to detect real-time chatter, and they demonstrated that hybrid signal processing methods are effective in enhancing the stability of machining [15]. Even with such improvements, it is a difficult challenge to have an overall correlation between machining parameters, dynamic response, and productivity measures, but at least where the structural dynamics evolve continuously during the process of material removal.

Chatter in thin-walled milling is a self-excited vibration phenomenon that is largely controlled by the regenerative effect whereby interaction between the current and previously generated surface modulate cutting forces and destabilize the machining process. This instability is further enhanced by the ever-changing stiffness and modal characteristics as the material is removed, resulting in nonlinear dynamic behaviour, and decreasing process stability [16]. Active chip thickness variation has a significant effect on force regeneration leading to amplified vibration amplitudes that worsen surface finish and dimensional accuracy and limit the rate of material removal that can be attained (MRR) [17]. It has been established in several studies that the stability of milling activities is highly influenced by the spindle speed, axial and radial depth of cut and the system dynamics, which necessitates the use of stability lobe diagrams and analytical models that are used to predict chatter [18].

Though useful in rigid systems, these models can be experimented with to fit thin-walled components, as the structural properties of these components are time-dependent [19]. Furthermore, the trade-off between maximizing MRR and keeping the system in the chatter-free zone is still a key issue, as increased cutting parameters are more likely to push the system to the unstable areas. Therefore, recent studies have aimed at combination of dynamic modelling with experimental validation to determine optimal cutting conditions that balance productivity and stability in milling of thin walls [20]. A study by Altintas et al. indicated that it is critical to correctly predict the location of stability lobes to optimize cutting parameters and prevent chatter in flexible milling systems [21]. Tyler et al. identified process damping, tools, geometry, and the stable cutting regions, especially in low-speed regimes [22]. Tamas et al. have emphasized the importance of experimental validation of stability diagrams, when considering thin-walled structures because of their changing dynamics [23]. Smith et al. demonstrated that sensor-based monitoring and signal processing methods are very effective in chatter detection and control [24]. According to Liu et al., the flexibility of the workpiece and the conditions of the fixturing are the key factors that determine both the amplitude of vibration and the reachable MRR [25].

Campa et al. showed that tool paths can be optimized to minimize deflection and improve machining stability in thin

floors [26]. Minis et al. contributed some early research on regenerative chatter mechanisms that are still used in the modern study of stability [27]. Sun et al. examined frequency response functional (FRF)-based techniques of enhancing the accuracy of chatter prediction [28]. The nonlinear dynamic effects in milling and how they influence stability limits were investigated by Mann et al. [29]. Balachandran et al. examined the effect of structural compliance on chatter onset, which confirmed that dynamic stiffness was dominant in determining the machining stable zones [30]. All these studies point out that high MRR in thin-walled milling can be achieved through a synergistic method of combining process optimization, dynamic modelling, and real-time monitoring to reduce chatter and to maintain productive machining. The Taguchi-based design of experiments (DOE) has emerged as a very popular methodology to optimize systematic parameters in milling processes due to its strength, less experimental effort being needed, and the ability to disarm the multi-factor interactions. When chatter instability and productivity (MRR) are conflicting goals, as often happens in the context of thin-walled milling, Taguchi approaches can offer a systematic means to determine the best machining conditions with limited trials [32]. Using orthogonal arrays like L27, statistical quantification of the effects of key process parameters, such as spindle speed, feed rate, and depth of cut, on response variables can be statistically quantified using signal-to-noise (S/N) ratios and analysis of variance (ANOVA) [33].

Recent research has established the fact that a combination of Taguchi and optimization and vibration-based response measurements can significantly enhance the accuracy of chatter characterization and control in machining processes [34]. Specifically, the inclusion of more advanced statistical measures like kurtosis, and the more traditional measures, such as RMS vibration, allow more sensitive detection of unstable cutting conditions. In addition, techniques of time-frequency analysis of the vibration behaviour such as Fast Fourier Transform (FFT) and wavelet-based methods have been increasingly used to capture the transient and non-stationary vibration behaviour as is common in thin-walled structures due to their dynamic nature. Such methods help to identify the predominant chatter frequencies and how they relate to machining parameters to enable a more holistic approach to optimization, providing a balance between stability and productivity [35].

Li et al. showed the applicability of hybrid Taguchi-grey relational analysis in the multi-response optimization of complex milling processes through a combination of multiple performance characteristics including surface roughness and MRR in milling processes [36]. Teti et al. have pointed out the need to integrate sensor-based monitoring and state-of-the-art signal processing to real-time assess machining performance, complementing Taguchi-based offline optimization [37]. According to Navarro et al., a combination of statistical design techniques with a frequency-domain feature can enhance the accuracy of chatter detection and the stability of the process [38]. Zawada et al. focused on the effect of parameters of a process on deformation of thin-walled components under the influence of vibration and concluded that it is necessary to optimize the choice of parameters of a process to achieve minimal errors under the influence of vibration [39]. Stavropoulos et al. have shown how the use of experimental design approaches to optimize cutting

strategies can be used to effectively suppress chatter and still achieve high MRR [40]. Mayr et al. emphasized the significance of integrating digital signal processing and machine learning in the present-day machining optimization frameworks [41].

Liu et al. demonstrated that time-frequency domain features yield the best sensitivity in detecting chatter as compared to traditional time-domain metrics alone [42]. Putra et al. discussed the multi-objective optimization methods to machining of advanced materials and highlighted the importance of statistical methods in obtaining process robustness [43]. Chen et al. evaluated the influence of machining parameters on vibration and productivity and showed the effectiveness of Taguchi-based methods to determine the optimal cutting conditions [44]. As shown by Singh et al., the joint application of Taguchi optimization and vibration analysis results in substantial improvements in the machining stability and performance in the process of milling [45]. All these studies combine to present a strong and viable model of chatter control and enhancement of material removal efficiency in milling thin-walled components using Taguchi based optimization, when combined with time frequency signal analysis.

Recent literature always determines that the time-varying structural dynamics that govern the chatter of thin-walled milling include regenerative effects, process damping, and changing stiffness that play a significant role in determining machining stability and surface integrity. There have been widespread uses of analytical and numerical models, such as stability lobe diagrams and frequency response-based approaches, to predict chatter onset, whilst experiments have focused on the predominance of spindle speed, feed rate, and depth of cut in controlling vibration behaviour. Signal processing approaches, in parallel, have enhanced the detection and description of chatter in non-stationary conditions: signal processing techniques, especially time-frequency analysis, higher-order statistical measures, such as kurtosis, and sensor-based monitoring. Maximization techniques in particular Taguchi and hybrid multi-response methods have been effectively used to optimize machining performance, balancing measures of productivity such as material removal rate (MRR) and quality measures such as surface roughness and amplitude of vibration. Most current studies however treat chatter indicators and productivity responses separately or based on single domain analysis with little emphasis on multi-axis vibration behaviour and how this can be directly integrated into a single optimization framework.

In this context, the present study introduces a novel experimental framework that integrates Taguchi L27-based multi-response optimization with time-frequency domain analysis of multi-axis chatter signals (X, Y, Z) for thin-walled milling of AA7075-T6. Unlike prior works, this research simultaneously incorporates material removal rate, kurtosis index as a higher-order chatter descriptor, and axis-specific chatter frequency into a single optimization and interpretation platform. The novelty lies in establishing a coupled RMS-kurtosis-frequency characterization linked with material removal dynamics, enabling a more sensitive and physically meaningful identification of stable cutting conditions under evolving structural stiffness. Furthermore, the study provides a comprehensive correlation between mass removal-induced dynamic variation and chatter behaviour, offering a unified perspective that bridges productivity and stability. This integrated methodology

advances beyond conventional single-metric or single-domain approaches and delivers a practical, experimentally validated strategy for achieving chatter-resilient, high-efficiency machining of thin-walled components.

2.0 MATERIALS AND METHODS

2.1 Material and Tool Specifications

The experimental research studies were conducted using AA7075-T6 aluminium alloy thin plates. Workpieces were made up of aluminium sheets of 40 x 80 x 3 mm thickness, in a cantilever-like configuration used to machine the thin-wall. Chemical composition of the material was determined with the help of spectrometric analysis, which confirmed that the major alloying elements include Zn (~5.58%), Mg (~2.53%), Cu (~1.68%), and Cr (~0.22%), which provide the necessary mechanical strength and machining characteristics. The milling experiments were performed on a Vertical Machining Center (AMS MCV 300) with a high-precision spindle and guideways with linear motion allowing the stable machining conditions and the accurate vibration analysis.

Table 1. Machine, Tool, and Material Specifications

Workpiece	Material	AA7075-T6 Aluminium Alloy
	Dimensions	40 × 80 × 3 mm (thin plate)
	Zn (%)	5.582
	Mg (%)	2.537
	Cu (%)	1.685
	Cr (%)	0.223
	Al (%)	~89.26
Machine	Machine Type	Vertical Machining Center (AMS MCV 300)
	Spindle Speed	60 – 6000 rpm (up to 10,000 rpm optional)
	Spindle Taper	BT-40
	Table Size	650 × 350 mm
	CNC Controller	Fanuc Oi-MF
	Rapid Traverse	40 m/min (X, Y, Z)
Cutting Tool	Tool Type	Solid Carbide Square End Mill
	Diameter	10 mm
	Number of Flutes	3
	Helix Angle	45°
	Rake Angle	10°
	Overall Length	80 mm

The machine has a spindle speed range up to 6000 rpm (extensible to 10,000 rpm), with a BT-40 spindle interface and CNC control (Fanuc Oi-MF), which ensures repeatability and stability of the process. Cutting tool used was a solid carbide square end mill having a diameter of 10 mm, three flutes, 45° helix angle and 10° rake angle, which was chosen to provide effective chip evacuation and low cutting forces during thin wall milling operation.

Figure 1 (A) shows the ready AA7075-T6 thin plate specimens that were prepared before machining, demonstrating that the sample dimensions (thin plates) were uniform and used in the experimental trials. The Vertical Machining Center (AMS MCV 300) used in this research is depicted in figure 1 (B), along with its CNC control interface and machining workspace. A detailed view of the spindle–tool assembly demonstrates the solid carbide end mill setup used for milling operations. Also, the fixturing setup of the workpiece on the machine table is shown in figure 1 (C), which will provide rigidity and repeatability when conducting chatter analysis experiments. Milling experiments carried out under well controlled laboratory conditions using instrumented setup to monitor the vibration in real time and record the required data. The re-creation of a set of thin-wall boundary conditions was achieved when a precision vice was used to mount a rigid clamped workpiece on the machine bed, while the dynamic behaviour of the material removal process was deemed to be captured during the removal process. It should be noted that the piezoelectric accelerometer was bonded to the surface of the working piece at an appropriate position to obtain the vibration signal in the three orthogonal directions of the working piece (X axis, Y axis and Z axis).

The signals received were then fed into a data acquisition (DAQ) system and stored periodically during each machining test. Sampling frequency has been selected to be sufficiently high to allow this sampling to fully reproduce the frequency components produced by chatter without aliasing. Vibration signals were then analysed in DewesoftX 2025.2 software for the time domain characteristics such as RMS, kurtosis, and frequency domain characteristics in the form of FFT analysis. Experimental conditions were observed to ensure that the measured data were repeatable and reliable to optimize and analyse the measured data later.

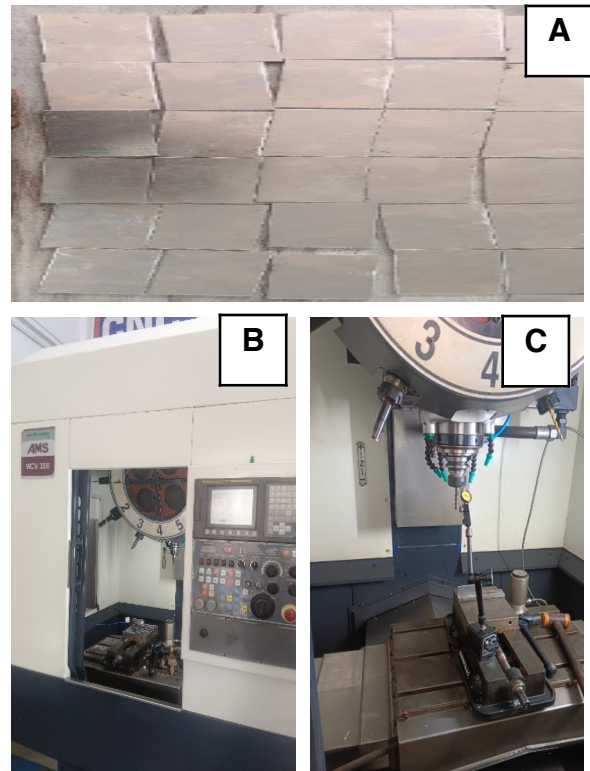


Figure 1. A) Machining Samples, B) Milling Machine,

C) Closeup View of Milling setup

Figure 2 depicts experimental layout adopted in chatter analysis of vibrations during milling. It depicts the CNC control interface, placement of accelerometer on the workpiece, and the integrated machine-DAQ system. This setup enables real-time acquisition and processing of machining-induced vibration signals.

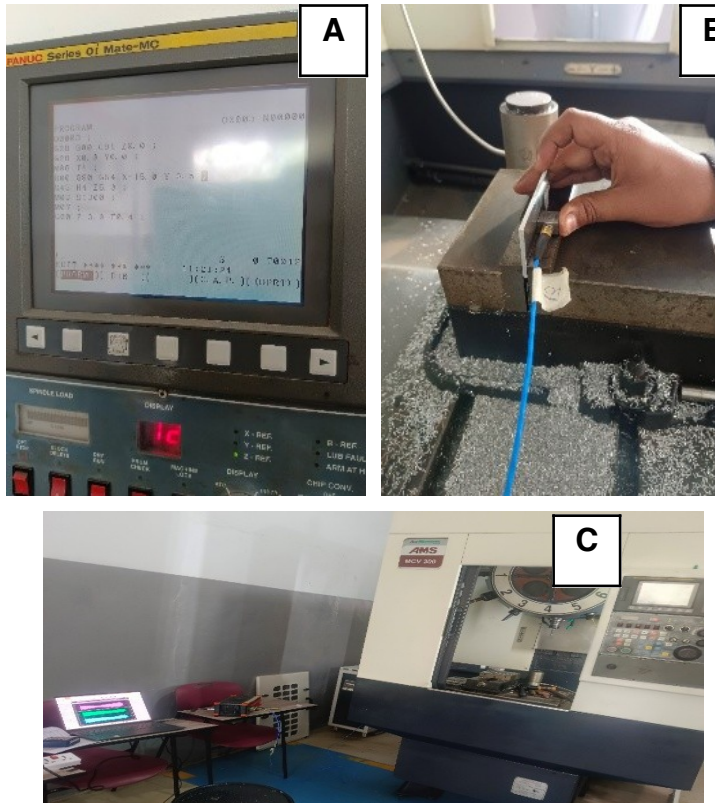


Figure 2 (A) CNC control panel, (B) Accelerometer mounted on workpiece, (C) Machine with data acquisition setup.

2.2 Design of Experiments (Taguchi L27 Orthogonal Array)

Taguchi structured design of experiments (DOE) method was used to systematically assess the effect of the machining parameters on the characteristics of chatter and material removal rate. A matrix designed for studying three control factors with three levels with minimal number of experimental trials was chosen, the L27 (3^3) orthogonal array. This design would provide equal (balanced) representation of factor-level combinations and allow efficient estimation of main effects, and less experimental effort would be required. The process variables used in each experimental run were varied, and the responses recorded were MRR, Kurtosis Index and chatter frequency (on the X/Y/Z axes). L27 array turns into a statistically robust design and then it may be analysed using signal to noise (S/N) ratios and ANOVA, which helps to determine the most conclusive factors that impact the machining performance.

2.3 Input Process Parameters and Levels

Input parameters chosen were those that had a substantial effect on the cutting mechanics and dynamism in milling operations. Three primary control factors have been considered, all at three levels, with consideration given to their nonlinear effects on the responses: spindle speed, feed rate and radial depth of cut. The levels were selected to provide a range of stable cutting conditions, but within a realistic operational range, due to machine capabilities, tool requirements and initial cutting tests. These parameters not only influence the chatter characteristics but also directly impact the chip formation, cutting forces, and vibration behaviour, which in turn makes its way to productivity.

Table 2. Input Process Parameters and Levels

Parameter	Symbol	Unit	Level 1	Level 2	Level 3
Spindle Speed	N	rpm	1500	2250	3000
Feed Rate	f	mm/min	100	125	150
Radial Depth of Cut	a _e	mm	0.5	1.0	1.5

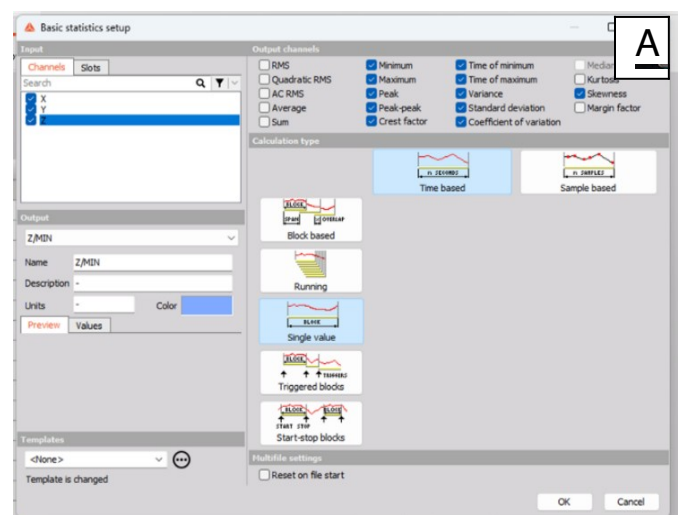


Figure 3 (A) Statistical setup for vibration signal analysis, (B) X, Y, and Z axis orientation for vibration measurement

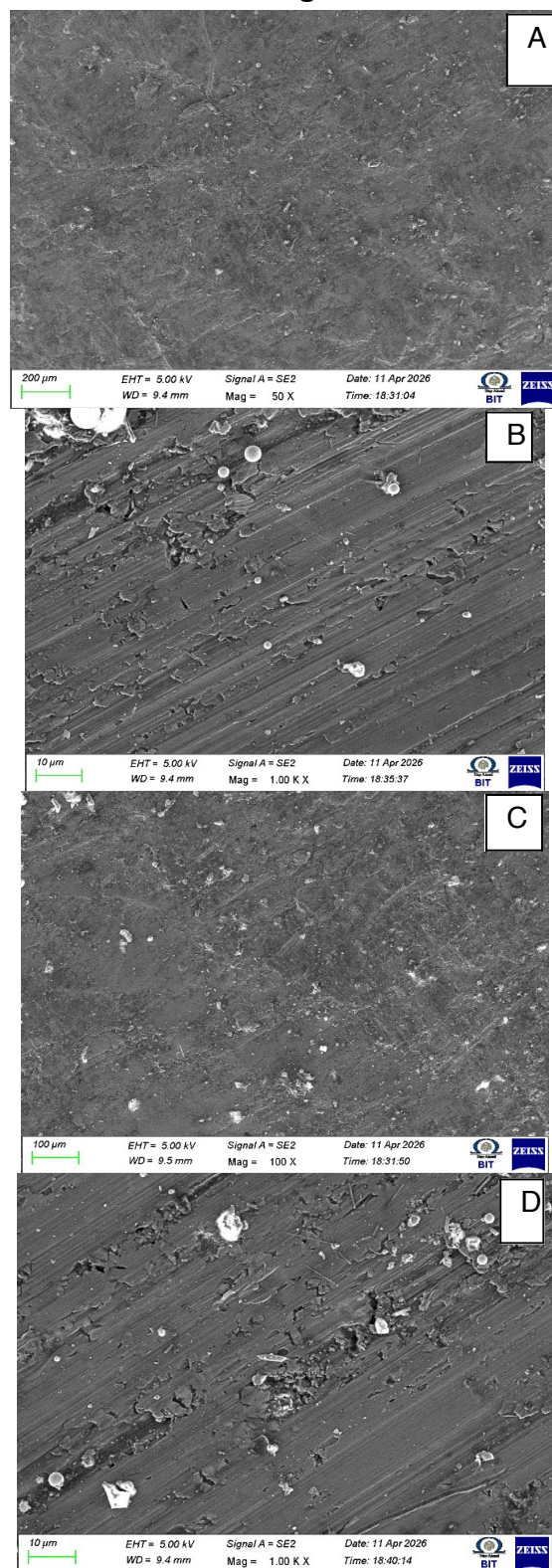
The methodology of vibration data acquisition and signal processing scheme employed for chatter analysis depicted in figure (3). The Extract vibration characteristics through statistical signal processing interface (Image (A)) is offered to extract vibration signals features like RMS, kurtosis in this phase. For machining induced vibration during milling operations, three orthogonal directions of vibration measurement were established, as presented in image (B), and the orientation of the accelerometer was determined considering these directions. The Taguchi signal-to-noise (S/N) ratio technique was used to measure the machining performance with respect to the desired response characteristics in the present study. The “larger-the-better” criterion was used for optimization of MRR which favoured higher material removal rate for better productivity. In contrast, the lower vibration severity or stability during machining indicated by the lower values of the two responses (kurtosis index and chatter frequency) was taken as the “smaller-the-better” criterion for the vibration characterization and analysis.

3. RESULTS AND DISCUSSION

3.1 Morphology and Elemental Analysis Before Machining

The FESEM morphological features of AA7075-T6 samples before cutting at varying magnifications are shown in figure 3, which demonstrates the primary surface appearance and the microstructure of AA7075-T6 before applying cutting force and dynamic loading. Micrographs showed that the surface had good material integrity prior to the machining process, in comparison with the other surface with irregularities and non-uniform distributed microstructural features. There is significant scatter of micro scale particulates on the surface, created by preparation and handling of the specimens, together with the-low level polishing scratches as imaged at the lower magnifications (A and C). No significant surface discontinuities, cracks or delamination features were observed, indicating the structural integrity of the material before milling. In larger magnifications, (Figure B and D) finer linear structures and shallower grooves start to form, likely resulting from initial surface preparation and cutting to create the specimen. The micrographs also shows the occurrence of uniformly distributed secondary phase particles dispersed in the aluminium matrix which indicates the attainment of proper alloy homogenisation during the heat-treatment process of the temper condition, T6. In addition, the lack of large voids or inclusions that are grouped indicates good metallurgical quality and less susceptibility to crack propagation locally during machining. Uniformity observed at the microstructural level appears of special significance for the thin-walled milling applications in which there is no chance to eliminate any other boundary conditions during the removal of material, which can contribute to the generation of chatter and the propagation of vibrations and dynamic instability. The relatively good surface quality of the material prior to machining provides a basis for comparing the machined

surface quality and allows for an accurate assessment of the damage caused by the chatter after machining.



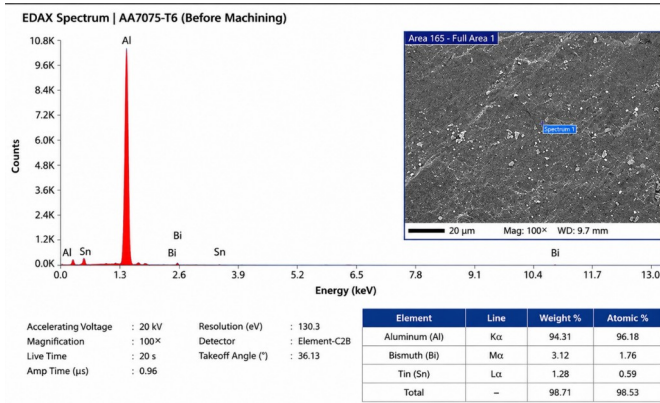


Figure 4: (A) FESEM image of unmachined surface at lower magnification, (B) Surface morphology at higher magnification, (C) Microstructural distribution, (D) Surface features and particle distribution, (E) EDAX spectrum and elemental analysis of AA7075-T6.

E-gam from the EDS spectrum and also elemental analysis illustrated in the figure 3(E) indicates that the elemental composition and chemical homogeneity of AA7075-T6 alloy employed in the present studies. This spectrum is mainly dominated by a very strong aluminium peak, proving that the main matrix phase of this alloy system is aluminium. Results of quantitative elemental analysis provide the aluminium-rich elemental composition with around 94.31wt. % aluminium and around 96.18 % atomic aluminium concentration. The other elements detected were Aluminium (Al) and Minor traces of bismuth (Bi) and Tin (Sn), weighted percentages of which were 3.12% and 1.28% respectively. Their existence may affect the local deformation behaviour, chip properties and tribological phenomena under machining. The elemental distribution across the analysed area is uniform, which means that the metallurgical consistency of the metal is satisfactory, and no sever segmentation effects and bean patching can be seen. Moreover, the nearly invisible state of the contaminating

peaks is indicative that no significant contamination occurred on the specimen surface, and thus the sample preparation protocol before FESEM characterization has been reliable. Aluminium is a major element in the alloy, part of its lightweight properties and is also a key part to adding the strength of precipitation hardening in the T6 temper condition, while the other alloy components are also important as a part of the strengthening mechanisms. The identified elemental composition is especially pertinent to the machining aspect in view of the fact that during the thin wall machining of the metal, the characteristics of the alloy chemistry shows a direct influence to the generation of driving cutting force, the characteristics of the thermal softening of the metal, the tendency of the metal to adhere to the tool, and the susceptibility of chatter phenomenon. EDS results are consequently used to validate the composition for correlation with the experimentally observed vibration, stability, and surface integrity responses for the milling experiments and the material's behaviour in its microstructure.

3.2 Taguchi-Based Optimization of Material Removal Rates

3.2.1 Experimental Results for MRR

From the experimental results, it is found that spindle speeds, feed rates, and radial depth of cuts are significant factors in material removal rate in the thin-walled milling of AA7075-T6 using Taguchi L27 experimental design. In all selected parameters, an increase of spindle velocity had its direct effect on the machining productivity, wherein an increase in spindle velocity has increased the cutting engagement and chip removal efficiency thus increasing the MRR. Insufficient or too high rotation speeds, however, can also lead to increased vibration excitation at the tool-workpiece interface, thus intensifying dynamic instability.

Table 3. Experimental Design Matrix and Measured Response Parameters Based on Taguchi L27 Orthogonal Array

S. No	Spindle / tool Speed (rpm)	Feed rate (mm/min)	Radial Depth of Cut (mm)	Chatter Frequency (Hz)			Kurtosis Index			MRR (mm ³ /min)	S/N Ratio	Mean
1	1500	100	0.5	75.07	75.07	75.07	14.27	6.79	6.64	150	43.5218	150.00
2	1500	100	1	75.07	75.07	75.07	26.04	8.90	9.81	300	49.5424	300.00
3	1500	100	1.5	75.07	75.07	75.07	120.56	16.74	14.40	450	53.0643	450.00
4	1500	125	0.5	75.07	75.07	75.07	8.78	4.39	4.86	187.5	45.4600	187.50
5	1500	125	1	75.07	75.07	75.07	26.86	13.52	12.34	375	51.4806	375.00
6	1500	125	1.5	75.07	75.07	75.07	11.90	18.85	14.89	562.5	55.0025	562.50
7	1500	150	0.5	75.07	75.07	75.07	12.77	6.33	6.00	225	47.043	225.0

											7	0
8	1500	150	1	75.07	75.07	75.07	18.33	16.5 4	16.6 3	450	53.064 3	450.0 0
9	1500	150	1.5	75.07	75.07	75.07	15.64	14.6 6	17.7 5	675	56.586 1	675.0 0
10	2250	100	0.5	112.3 0	112.3 0	112.3 0	7.08	5.86	5.86	150	44.383 7	168.7 5
11	2250	100	1	112.3 0	112.3 0	112.3 0	8.75	9.87	12.9 2	300	49.542 4	300.0 0
12	2250	100	1.5	112.3 0	112.3 0	112.3 0	17.80	49.4 2	40.7 3	450	53.064 3	450.0 0
13	2250	125	0.5	112.3 0	112.3 0	112.3 0	9.87	6.52	6.55	187.5	46.042 0	191.5 6
14	2250	125	1	112.3 0	112.3 0	112.3 0	10.61	10.4 9	11.6 5	375	51.480 6	375.0 0
15	2250	125	1.5	112.3 0	112.3 0	112.3 0	14.09	10.7 2	10.5 8	562.5	55.002 5	562.5 0
16	2250	150	0.5	112.3 0	112.3 0	112.3 0	9.69	7.06	7.29	225	47.043 7	225.0 0
17	2250	150	1	112.3 0	112.3 0	112.3 0	14.52	10.6 2	11.3 1	450	53.064 3	450.0 0
18	2250	150	1.5	112.3 0	112.3 0	112.3 0	10.99	10.1 7	10.2 3	675	56.586 1	675.0 0
19	3000	100	0.5	150.1 5	150.1 5	150.1 5	31.63	45.9 8	42.4 8	150	43.521 8	150.0 0
20	3000	100	1	150.1 5	150.1 5	150.1 5	12.35	16.8 6	15.3 6	300	49.542 4	300.0 0
21	3000	100	1.5	150.1 5	150.1 5	150.1 5	11.45	8.67	8.96	450	53.064 3	450.0 0
22	3000	125	0.5	150.1 5	150.1 5	150.1 5	25.28	36.1 1	21.9 6	187.5	45.460 0	187.5 0
23	3000	125	1	150.1 5	150.1 5	150.1 5	9.23	8.41	8.91	375	51.480 6	375.0 0
24	3000	125	1.5	150.1 5	150.1 5	150.1 5	7.54	9.99	10.1 2	562.5	55.002 5	562.5 0
25	3000	150	0.5	150.1 5	150.1 5	150.1 5	29.63	45.9 1	47.4 3	225	47.043 7	225.0 0
26	3000	150	1	150.1 5	150.1 5	150.1 5	6.83	7.55	8.06	450	53.064 3	450.0 0
27	3000	150	1.5	150.1 5	150.1 5	150.1 5	9.32	7.34	7.56	675	56.586 1	675.0 0

Feed rate was found to have a positive linear correlation with MRR as the feed rate became greater the amount of material removed was greater in some way, due to increased chip load. However, high feed conditions have been shown to also increase cutting forces and risk of chatter starting, especially when operating in a thin wall configuration, which is less structurally stiff. Enhancement of MRR was found to be highly dependent on radial depth of cut as larger engagement between the cutting tool and the workpiece reached a significant level of contribution towards

the enhancement of MRR value. Interaction area between the tool / the workpiece grew significantly at higher radial depths, thus causing higher cutting forces and the new dynamics. These all interact with each other, and determine the trade-off between acceleration and process stability, making it imperative to carefully balance the cutting conditions to maximize MRR with achieving chatter-free production during milling operation

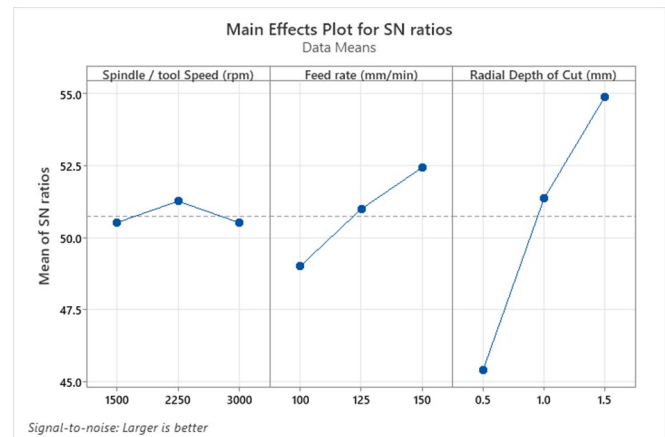
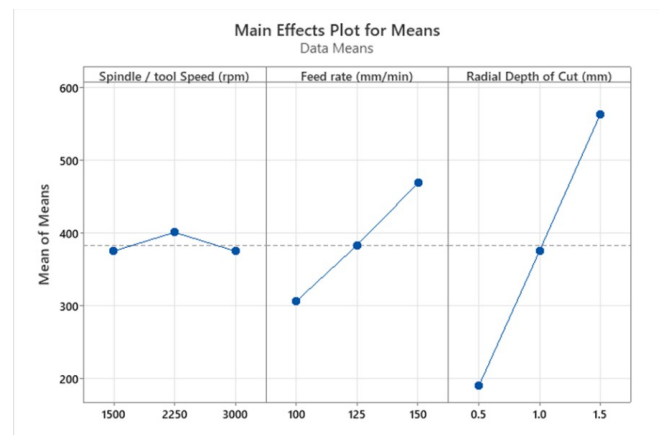
Table 4. Response table for signal to noise ratios

level	Spindle / tool Speed (rpm)	Feed rate (mm/min)	Radial Depth of Cut (mm)
1	50.23	49.02	45.43
2	51.27	51.00	51.36
3	50.53	52.44	54.88
Delta	0.74	3.42	9.45
Rank	3	2	1

Table 5. Response table for signal to Means

level	Spindle / tool Speed (rpm)	Feed rate (mm/min)	Radial Depth of Cut (mm)
1	375.0	306.3	189.8
2	400.8	383.3	375.0
3	375.0	468.8	562.5
Delta	25.8	162.5	372.7
Rank	3	2	1

The values of response that were generated from the Taguchi analysis are given in Tables 4 and 5 for the signal-to-noise (S/N) ratio and mean MRR values, respectively. Delta values and ranking on every step definitely give the idea about that the most influential parameters with respect to the MRR are the feeding rate, spindle frequency and depth of cut. Remarkably higher delta value of radial depth of cut shows its predominant contribution to enhancing productivity of the machining process. In addition, both S/N ratio and the mean response ranking indicates the same tendency, which indicates that the experimental data and Taguchi optimization is reliable

**Figure 5.** Main effects plot for S/N ratios of MRR**Figure 6.** Main effects plot for mean MRR values under different machining parameters

Figures 5 and 6 shows the main effects plots of the S/N ratio and MRR values for different machining conditions, respectively. It is observed from the plots that MRR has relatively high significance with feed rate and radial depth of cut as compared to the spindle speed. Increase in the steep slope of the radial depth of cut reflects that it has a significant influence on the material removal performance. The spindle speed of 2250 rpm, feed rate of 150 mm/min and radial depth of cut of 1.5 mm were selected as the optimal machining parameters for maximum MRR from the result of S/N ratio analysis.

3.2.2 Regression analysis

A multiple linear regression analysis is conducted to build relation between the machining parameters and material removal rate (MRR) for thin-walled milling of AA7075-T6. A maximum of four input variables, namely spindle speed, feed rate, radial depth of cut and tool material, were selected for this study and MRR was employed as the output variable. The regression model that can be produced from the experimental data is

$$MRR = -375 + (8.5 \times 10^{-14} \times \text{Speed}) + (3 \times \text{Feed}) + (375 \times \text{RDOC})$$

where:

Speed = spindle speed (rpm)

Feed = feed rate (mm/min)

Radial Depth of Cut = radial engagement (mm)

The regression model had coefficient of determination (R²) value of 97.8% and an adjusted R² value of 97.5% showing high correlation between the modelled and experimental MRR. The statistical results from the ANOVA analysis obtained that feed rate and radial depth of cut significantly affected the MRR with $p < 5\%$ while spindle speed had relatively low statistical significance. From the parameters selected found the most significant contribution to MRR enhancement was due to radial depth of cut which increased the surface area of engagement between the tool and the workpiece.

Table 6 Regression Coefficients for MRR Model

Parameter	Coefficient	Standard Error	t-Value	p-Value
Constant	-375.000	39.700	-9.446	0.000
Spindle Speed	0.000	0.009	0.000	1.000
Feed Rate	3.000	0.255	11.747	0.000
Radial Depth of Cut	375.000	12.769	29.368	0.000

3.2.3 Analysis of Variance (ANOVA) for MRR

Thin Wall Milling of AA7075-T6 was used to analyse the effect of the machining parameters on the material removal rate using Analysis of variance (ANOVA). The analyses were performed at 95% confidence level to assess the contribution of spindle speed, feed rate and radial depth of cut for MRR variation. The ANOVA results indicate that feed rate and radial depth of cut have significant statistical impact on MRR at $p < 0.05$ while spindle speed has a negligible statistical contribution at the chosen experimental range for MRR. The F-value for radial depth of cut was the highest and percentage contribution among all parameters, it confirms the importance of this influence on the productivity of the chip formation process. This behaviour has been attributed to an increase in cutting engagement area and a greater volume of chips removal with the larger radial depths. Feed rate was also found to have significant effect as it increased the chip load, chip removal rate per unit time. However, the effect of spindle speed on MRR was negligible since a large range of spindle speed was used which mainly affected the cutting dynamics, and the MRR was not one of the major parameters. The results from ANOVA match the Taguchi response analysis and main effects plots, thus confirming the validity of the experimental design and optimization approach.

Table 8 ANOVA Results for Material Removal Rate (MRR)

Source	DF	Sum of Squares (SS)	Mean Square (MS)	F-Value	P-Value
Spindle Speed	2	0.00	0.00	0.00	1.00

Feed Rate	2	101250.00	50625.00	60.00	0.00
Radial Depth of Cut	2	632812.50	316406.25	375.00	0.00
Error	20	16875.00	843.75	-	-
Total	26	750937.50	-	-	-

ANOVA analysis showed that the depth of cut is the most significant parameter, followed by the feed rate and spindle speed in affecting the MRR in the material removal rate. Parameters with $P < 0.05$ are statistically significant towards variation in MRR. The selected factors, which are the factors with significant contribution, are the radial depth of cut which is due to the increase of tool – workpiece contact during machining and the increase of the volume of chip removal. Likewise, productivity was significant different for feed rate because it results in more material cut in less time due to increased chip load. Contrastively, within the testing range, spindle speed had relatively small influence on volumetric material removal. These observations are satisfactory as compared to the Taguchi Response Analysis and Main effects plot.

3.2.3 Confirmation Experiment

In order to validate optimal machining parameters from Taguchi optimization analysis, a confirmation experiment was done. The optimum parameter combination that maximized material removal rate was determined as spindle speed of 2250 rpm, feed rate of 150 mm/min and radial depth of cut of 1.5 mm, using signal-to-noise (S/N) ratio analysis. These were the optimal conditions used for the confirmation test and the experimental MRR was compared with the estimated MRR using the Taguchi model. Derived results showed a good match between theoretical and experimental results, verifying the validity and reliability of the formulated optimization method. Percentage error calculated between the calculated and measured values is very small, which means that the successfully obtained low percentage error will lead to the conclusion that the Taguchi methodology is able to predict the machining performance within the selected operating range. Optimum condition MRR was predicted using:

$$\hat{\eta}_{MRR} = \eta_m + \sum_{i=1}^n (\eta_i - \eta_m)$$

where:

- η_{MRR} = predicted MRR at optimal condition
- η_m = overall mean MRR
- η_i = mean response at optimal level of each factor
- η = number of significant parameters

• Table 9. Experimental confirmation

Response	Predicted Value	Experimental Value	Error (%)
MRR (mm ³ /min)	675.0	668.4	0.98%

The optimal machining parameters from the Taguchi S/N ratio analysis were used in the confirmation experiments to validate the developed optimization model for material removal rate (MRR). The MRR value was measured experimentally as 668.4 mm³/min while the MRR value was predicted from the Taguchi estimation equation as 675.0 mm³/min and prediction error were only 0.98%. A good agreement between the predicted and experimental responses supports the appropriateness, reliability, and predictive nature of the proposed optimization methodology in milling thin-walled AA7075-T6 material.

4. Kurtosis Analysis

Four thin-walled milling processes were performed to obtain the variation of the kurtosis index along the X, Y, and Z directions under different milling processes during thin-walled milling of AA7075-T6 is shown in figure 7. Kurtosis is a higher-order statistical characteristic that reflects the impulsive characteristics and irregularity of vibration signals and is used to track the severity of chatter and the degree of dynamic instability during the machining process, thus serving as a good indicator of chatter severity and dynamic instability. The results indicate larger variations of the kurtosis values within the different experimental runs which show the different conditions of stability and instability for the various combinations of spindle speed, feed rate and radial depth of cut.

Optimum vibration performance is expected in the cutting regions characterized by lower kurtosis values as seen in several runs, while those zones with higher peaks are associated with chatter initiation and enhanced dynamic response, respectively. It was found that severe vibration bursts and intermittent instability was happening during material removal at prominent kurtosis peaks (at runs 3, 12, 19, 22 and 25). The Z axis always presented higher Kurtosis in all the directions measured than the X and Y axes, which indicates that in milling the thin-walled structure, the components of vibration were more dominant in the vertical direction than in the X and Y directions. As a result of the workpiece stiffness affected and cantilever-type deformation properties under cutting force, this behaviour is attributed.

Some moderate variations were also observed for the Y-axis and exhibited relatively smaller kurtosis values during most experimental runs for the X-axis. In addition, a higher increase in the higher level of machining parameters led to an increase in the kurtosis, implying that the higher the level of the cutting engagement and chip load parameters, the more intense the dynamic excitation and the chatter formation. The observed trends validate the advantage of the Kurtosis analysis as a sensitive and reliable tool to detect unstable machining conditions and assess chatter behaviour in the case of thin-walled manufacturing process.

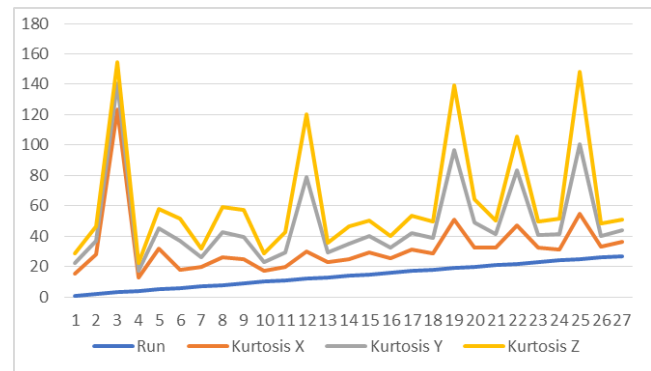


Figure 7. Variation of kurtosis index along X, Y, and Z directions under different machining conditions

5. Chatter Frequency Analysis

The frequency response of chatter is shown in figure 8 in the thin-walled milling experiments under various milling conditions. The conclusions revealed three predominant frequency levels that is, ~75.07 Hz, ~112.30 Hz and ~150.15 Hz for the spindle speeds selected. There was almost no change of frequency in the same parameter group, indicating that the dynamic excitation frequency mainly depends on the spindle speed during milling process. It was observed that at lower spindle speed conditions, the chatter frequency was present around 75 Hz, which shows relatively lower level of dynamic excitation and a stable cutting condition.

The chatter frequency gradually approached 112 Hz and 150 Hz which were associated with the increasing spindle speed (intermediate and higher), indicating the enhancement of tool-workpiece interaction frequency and the response of vibration. This trend suggests that the rotational speed has a direct effect on the regenerative chatter mechanism through the effect on the tooth passing frequency and the generation of dynamic force in material removal process. The progressive growth of chatter modes with spindle speed also reflects the growth of the thin-walled workpieces structural excitation point with spindle speed, where the stiffness is small, and structural vibration is easily excited. Furthermore, at higher cutting frequency values, resonance-like phenomenon can be expected in the machine which may lead to increasing of vibration amplitudes and surface roughness also at elevated cutting conditions.

Repeated vibration data was obtained in various experimental runs to validate the repeatability and reliability of the acquired vibration data. Furthermore, the frequency domain analysis is used to explore and supplement the time domain analysis using kurtosis to characterize the chatter-generating vibration bands. The sum of the frequency response and statistical vibration indications is substantiated by the fact that the chatter behaviour can be strongly exacerbated by implementing more aggressive cutting methods, such as cutting engagement and spindle speed increase, in thin-walled manufacturing process. Hence, chatter frequency analysis can be used to deduce the dynamic stability properties of the whole machining process and helps to detect the stable working range for better machining performance and surface integrity.



Figure 8. Chatter frequency response obtained during thin-walled milling experiments.

6. Morphology of the Milled Samples

The FESEM surface image with respect to each milling condition to study the surface morphology changes of the AA7075-T6 surfaces under various conditions can be found in figure 9. Figure 9 shows the surface morphology of machined AA7075-T6 surfaces for various milling conditions, and stable operation seems to be linked to a particular surface topography. The FESEM surface morphology under different milling conditions is as shown in figure 9, which shows that different surface morphologies are associated with different machining stability and chatter behaviour. The micrographs clearly revealed of the influence of dynamic and process parameters on the surface integrity, material plasticisation, nature and interaction of the chips in the process of thin wall parts fabrication by milling. In comparison the surface morphology of image A is relatively smooth with comparatively regular feed marks and fewer signs of surface damage suggesting a relatively stable cutting situation with lesser severity of vibration. However, machining grooves can be seen here, together with debris sticking and even slightly more pull-out of material in both the direction of the cut, and in areas, suggesting greater frictional interactions and the onset of instability due to chatter. Severe surface degradation, with irregular feed patterns, plastic deformation zones, smeared material zones, broken chip adhesion and micro-scale tears of the machined surface is shown again in the higher magnification pictures of C and D.

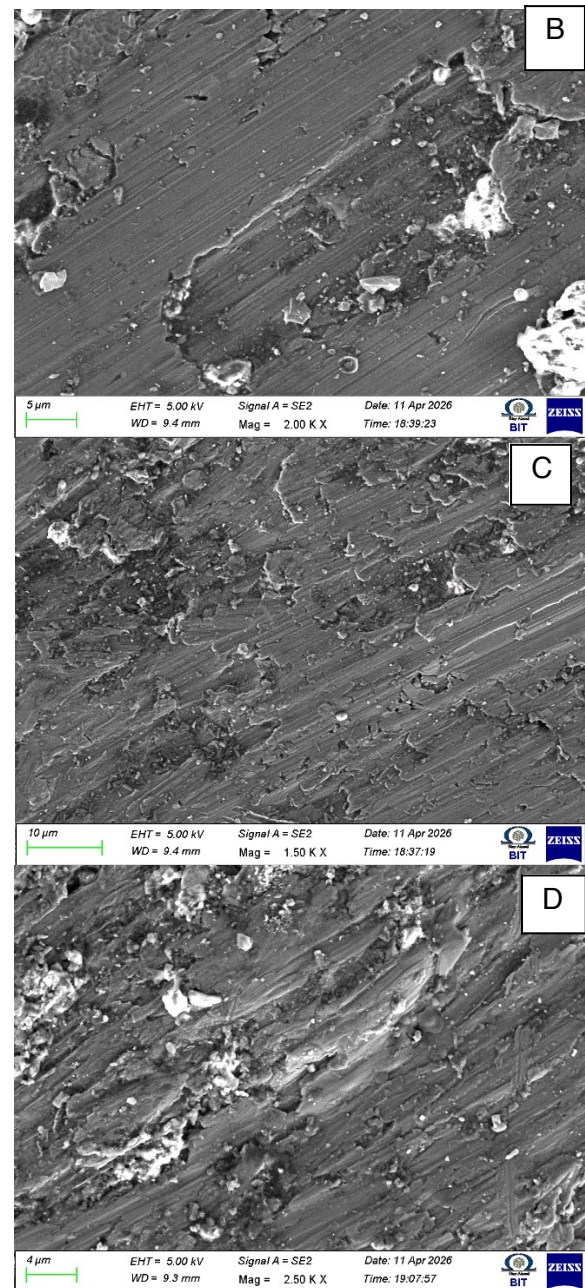
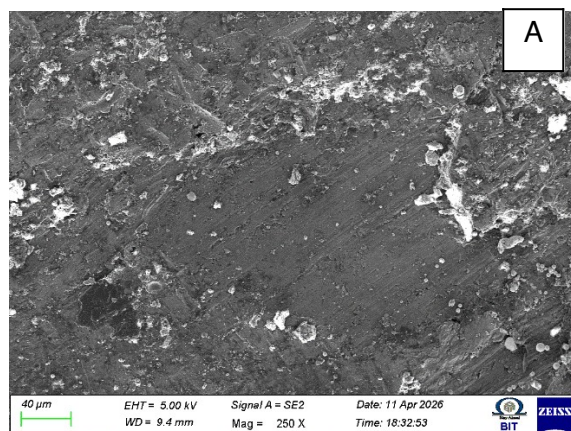


Figure 9. FESEM morphology of machined AA7075-T6 surfaces under different milling conditions.

These features are a result of varying cutting force conditions and the effect of vibrations on the cutting tool which underlies regenerative vibration effects during material removal, resulting in unstable machining behaviour. Sticking Particles and discontinuous grooves also indicate the possibility of intermittent chip uptake and of higher degree of tool-workpiece interaction at aggressive cutting conditions. Moreover, as shown by a good correlation, the observed non-uniform surface texture and surface waviness are also related to the increased kurtosis values, and higher frequencies measured by the vibration analysis, showing the significant correlation between dynamic instability and surface aspects in thin-walled milling operations. In the analysed micrographs, the worst case presents in image D, where both the plastic flow and smearing of the material is intensified on the surface and also irregular; these

characteristics suggest that the chatter has been excited significant, and the machining processes are not so stable as the others. In conclusion, the findings of the FESEM observations confirm the results obtained from the chatter characterization in vibration mode and confirm that the machining conditions can be considered unstable when the cutting engagement is increased, thus having a significant negative effect on surface integrity of the AA7075-T6 thin-walled component.

7. Conclusions

The present study covers the experimental investigation of the chatter characteristics and material removal behaviour in thin-walled milling of aluminium alloy AA7075 using Taguchi orthogonal array L27 design. Time- and frequency-domain vibration characterization was used to investigate material removal rate (MRR), the kurtosis index and chatter frequency for different spindle speeds, feed rates and radial depth of cut. Based on the Taguchi optimization results, analysis of the S/N ratio and ANOVA results, this showed that the radial depth of cut is the most influence parameter on MRR followed by feed rate and spindle speed parameter. The optimum machining parameters for the maximum MRR were determined at 1500 rpm, 1.5 mm radial depth of cut and 150 mm/min feed rate and regression coefficient was found to be very close to the experimental responses showing good agreement between the predicted and experimental responses whereas the confirmation experiment showed that there is very little prediction error made with the optimization model so developed making it adequate and reliable.

Since the cutting conditions were not very stable, the kurtosis analysis method was found to be very effective and feasible because significant changes were observed under unstable cutting conditions, and the workpieces were thin walled, making the Z-direction have a relatively higher level of vibration severity caused by the relative lower structural stiffness. Frequency analysis showed that chatter frequency grew gradually with spindle speed which showed that at the higher spindle speeds, the dynamic excitation and the behaviour of regeneration of the chatter vibration increased. Additionally, the FESEM surface morphology image analysis of the manufactured surfaces validated the presence of surface deterioration: irregular feed marks, debris accumulation, plastic deformation, material smearing at unstable cutting conditions owing to chatter effect. Optimisation and vibration analysis and surface morphology interpretation are two methods that can be integrated into a robust framework for assessing the stability of thin-walled milling processes and enhancing productivity.

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